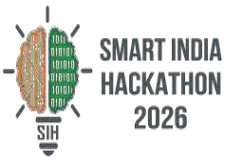


SMART INDIA HACKATHON 2026

DARPAN (SIH-DETA)



Self-Growing Hybrid AI & Railway Physics Engine for Coaching Train ETA

- Problem Statement ID – SIH26028
- Problem Statement Title – Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains
- Organization / Dept – Ministry of Railways (MIC)
- Theme – Smart Automation | Category – Software
- Team ID – 131076 | Team Name – Twarit



TEAM TWARIT — ENGINEERING DELIVERABLES

6-MEMBER SPLIT

Lead & Architect

2-Pass Hybrid Engine, Space-Time DAG & Zero-Clash Platform IntervalTree

AI / ML Lead

31-Feature Quantile GBDT (P10/P50/P90) & Delay-Jump Classifier (0.92 AUC)

Data & Scraper

1.68 GB 6-DB Data Lake, 24x7 Failover Scraper & Feedback Scorer Loop

Backend & API

FastAPI /v1/deta Gateway, <1.5ms Redis Cache & Manual Accident API

UI & Rail Ops

React 18 + Leaflet GIS (3 User Views), G&SR Fog/Heat Rules & 20K Backtest

1.68 GB (Self-Growing)
6-DB Live Scraping & Feedback Lake

8,990 Stns · 704 Cabins
417,080 Schedule Stops Mapped

1.51M Delays · 3.19M Wx
73,342 Track Recovery Profiles

4.93m → 1.49m (-70%)
Typical Error Cut (N=20,000 Test)

85.3% Window + Manual Ops
P10-P90 Band + Live Accident Input

GIST: Replaces Indian Railways' static **Schedule + Current Delay - Fixed Buffer** formula with a **Self-Growing Hybrid AI + Railway Physics Engine** that predicts **Best / Expected / Worst-Case arrival windows (P10/P50/P90)** & supports **live manual accident overrides**.



Core Idea (How It Works)

- 2-Step Hybrid Engine (AI + Rail Rules):** Step 1 predicts travel drift via **31-Feature Gradient Boosted AI [P10/P50/P90]**; Step 2 enforces track & platform physics [**Space-Time DAG**].
- Smart-Zoom Route Tracking:** Runs fast **station-to-station [0(1) Mode]** when on time; zooms into **halt-by-halt signal cabins [704 Cabins]** when delayed.
- Self-Growing DB & Feedback Flywheel:** Constantly scrapes live runs & scores its own past predictions [**1.68 GB now**]**—the longer it runs, the smarter & more accurate it gets.**
- Manual Unpredicted Accident Input:** Control rooms can inject **sudden unpredicted accidents/blocks [1-Click Override]** to instantly recompute ETAs & 25kV detours.



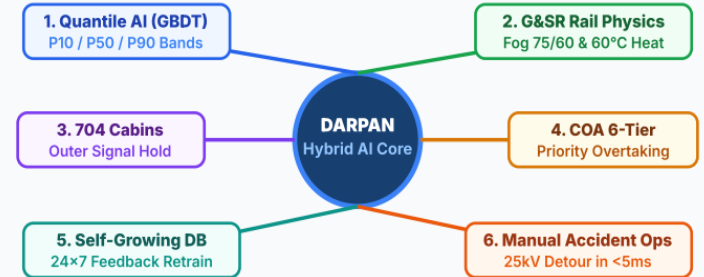
How We Solve Every PS Point

- Replaces Static Recovery Buffers:** CAG audits show fixed buffers fail on **>100% congested trunk lines**. We learn real section slack & enforce **no early departure before schedule [STD Hold]**.
- Overtaking & Following-Train Delays:** Models **Vande Bharat / Rajdhani priority overtaking [COA 6-Tier]** and **3–7 min safety headway gaps** behind slower trains.
- Official Fog, Rain & Track-Heat Rules:** Converts live weather into **Indian Railways G&SR Fog-Pass speed limits [75/60/30 km/h]** & **60°C rail-buckling** speed caps.
- Zero-Clash Platforms & Crowd Surges:** [**IntervalTree**] prevents platform clashes, predicts **Outer Home Signal wait times**, and models **Festival Crowd Halts (Maha Kumbh 3.2x)**.



Core Architecture Mindmap & Results

DARPAN SOLUTION MINDMAP



✓ **85.3% Window Accuracy & 0.92 AUC:** Calibrated [P10, P90] arrival window band + **Renamed Code Bridge** (+72.5K rows rescued).

Held-Out Error: Old Formula vs. DARPAN **-69.8% Error**

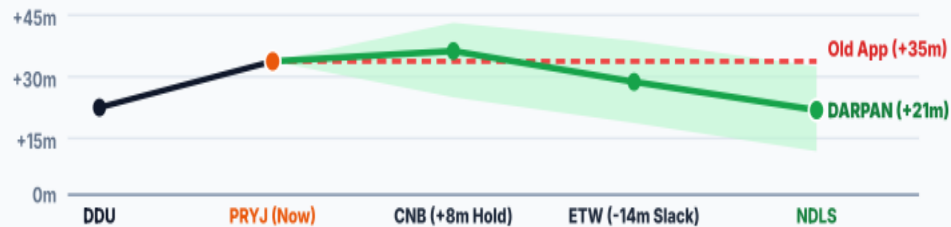
Old Formula (STA+Delay)	4.93 min
DARPAN AI + Physics	1.49 min (MedAE)

TECHNOLOGY STACK & TRAJECTORY FORECAST

- 24x7 Scraping & Weather Ingestion:** 4-Source Failover (ConfirmTkt→NTES→eRail) + Open-Meteo 180-Hex Grid + **Smart Pacing (-79.9% load)**.
- Self-Growing 6-Database Lake (1.68 GB+):** 6 SQLite DBs (darpan, historical, weather, stations, telemetry, anomalies) + **<1.5ms Cache**.
- Layer 1 — 31 G&SR Operational Features:** Speed trend, 73.3K slack profiles, **Fog limits (75/60/30 km/h)**, **60°C rail heat** & festival surges.
- Layer 2 — Arrival Window AI:** 3 Quantile GBDT models predict **P10 / P50 / P90** (Transit $R^2 = 0.987$) + **Delay-Jump Alert (0.92 AUC)**.
- Layer 3 — Physics & Manual Accident Ops:** Space-Time DAG + **Zero-Clash Platforms** + **1-Click Manual Overrides** + **25kV Detours**.
- Feedback Self-Scoring Flywheel:** SnapshotWriter + score_snapshots.py continuously audit predictions vs. actual arrivals to retrain AI.

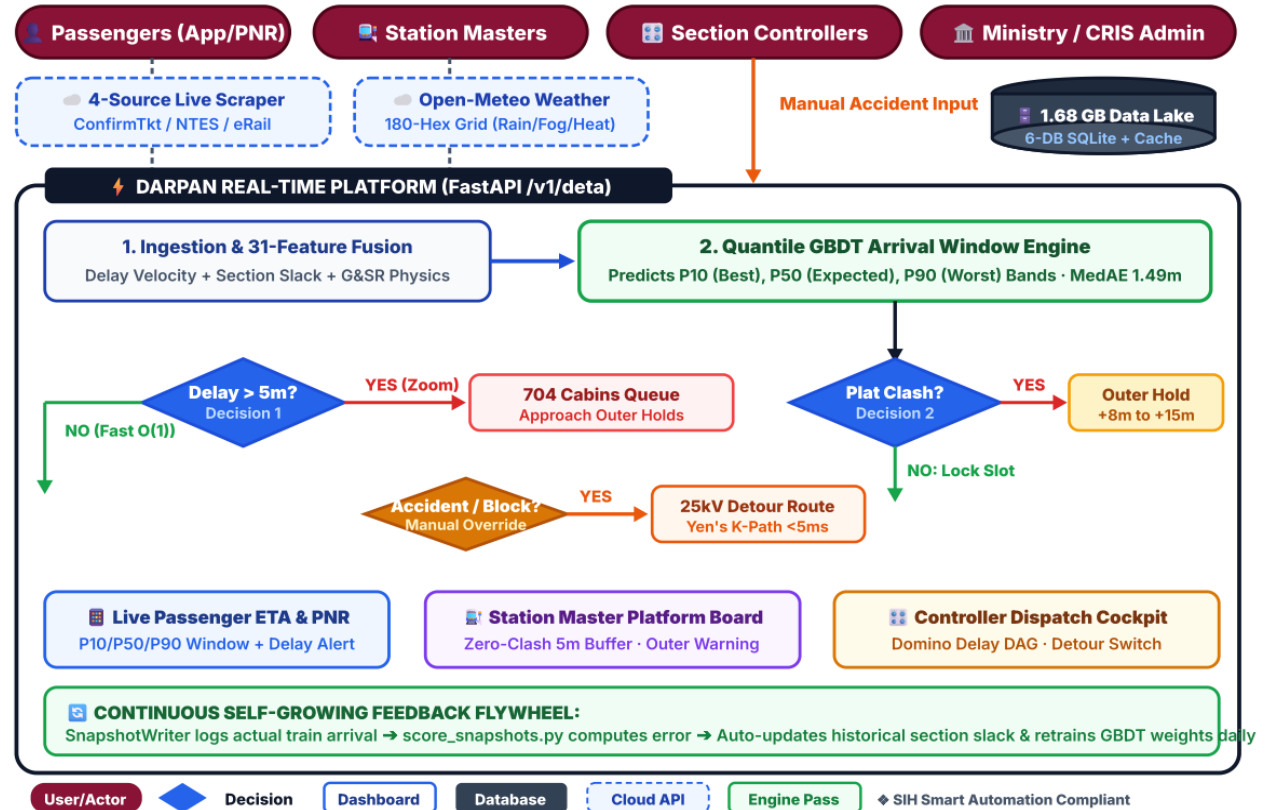
Live 12301 Rajdhani Trajectory (Old Line vs. P10-P90 Cone)

Slack & Fog



SYSTEM DATA FLOW DIAGRAM (DFD) — LEVEL 1/2

LIVE & REAL-TIME



PROCESS FLOW:

1. Multi-Source Scrape + Weather

2. 31-Feature G&SR Fusion

3. Quantile GBDT
(P10/50/90)4. Physics DAG + Manual
Ops

5. FastAPI (<18ms)

6. Self-Scoring DB Loop

1. FEASIBILITY & ECONOMIC VIABILITY

✓ Self-Growing Public Data Lake (1.68 GB+)

Built on **1.51M real delays** & **8,990 stations**; 24x7 scraper & feedback scorer grow the DB daily—**longer it runs, better it performs.**

✓ ₹0 Hardware Capex & <₹15K/mo Cloud Opex

Saves **₹120+ Cr in trackside/GPS hardware**; runs on standard CPU in **<18ms** (zero GPU cost) with **-79.9% scraper bandwidth.**

✓ High Railway Operational & Fuel ROI

Prevents outer-signal train stops (saving **150–200 kWh traction power / 35–50L diesel per stop**) & cuts junction platform delays.

 Empirical Backtest Scorecard (N = 20,000)

Travel Time Fit (R ²)		98.7%
Stations Geocoded		96.7%
Delay Fit (R ² Score)		93.1%
Window Accuracy		85.3%
Typical Error Cut		-69.8%

2. REAL RAILWAY CHALLENGES

⚠ Internal Railway GPS (RTIS/COA) is Closed

ISRO 30s locomotive GPS and track signal relays are private CRIS systems unavailable to external teams.



⚠ Extreme 12–24h Delays & No Early Departures

Heavy delay outliers skew simple averages, while early trains must wait at stations until Scheduled Departure (STD).



⚠ Domino Congestion & Platform Clashes

Standard AI predicts trains in isolation—putting two trains on Platform 1 at the exact same minute.



⚠ Unpredicted Accidents & Renamed Stations

Sudden derailments/accidents cannot be predicted from history alone; renamed stations (ALD→PRYJ) break data.



3. HOW DARPAN SOLVES THEM (1:1)

🛡 4-Source Public Failover + 704 Cabin Queue Math

Cascades across **ConfirmTkt/NTES/eRail** + predicts outer-signal wait times across **704 mapped Approach Cabins.**

🛡 P10/P50/P90 Window AI + Schedule Clamp

Asymmetric quantile AI clamped at **STD** cuts typical error from **4.93m to 1.49m (-70%)** with **85.3% window coverage.**

🛡 Zero-Clash Platform Assigner + Overtaking Rules

IntervalTree enforces 5-min platform gaps, **COA 6-Tier priority overtaking**, & following-train headway cascades.

🛡 Manual Accident Input + 25kV Detour + Code Bridge

1-Click Operator Accident Input (**POST /v1/deta/incidents**) triggers **25kV detours** in **<5ms**; rescues **72.5K alias rows.**

1. PASSENGERS (APP & PNR)

- ❖ Shows **Best (P10), Expected (P50), & Worst-Case (P90)** arrival times instead of a single guess.
- ❖ Displays exact **buffer time recovered (-14m)**, fog speed alerts, & privacy-safe 10-digit PNR tracking.

85.3% Window Accuracy

2. STATION MASTERS (PLATFORMS)

- ❖ **Zero-Clash Platform Assigner** enforces 5-min safety gaps & warns of **OUTER HOLD +12m** delays early.
- ❖ Predicts **Festival Crowd Dwell Surges** (Maha Kumbh 3.2x) for cleaning, crew & cab/bus planning.

704 Approach Cabins Mapped

3. CONTROL ROOMS (MANUAL OPS)

- ❖ **Manual Accident Override Input:** Operators inject unpredicted accidents (Derailment, Cattle Hit, Wire Snap).
- ❖ Instantly recomputes downstream domino ETAs & **25kV Electrified Detour Routes** (Yen's K-Path).

Manual Accident + 25kV Detour

4. SELF-LEARNING RAILWAY API

- ❖ Fast **/v1/deta/*** REST APIs + **24x7 Self-Growing DB** that scrapes runs & learns from prediction feedback.
- ❖ Tags every value with honest source (live/cache/schedule) at **<1.5ms cached latency** across 17 zones.

MedAE: 1.49m (-70% Error)

👥 SOCIAL IMPACT (2.4 Cr Daily Passengers)

Eliminates platform waiting anxiety & missed train connections; improves station crowd safety during festivals.

💰 ECONOMIC IMPACT (₹120+ Cr Capex Saved)

Boosts throughput on >100% congested trunk lines with zero new hardware; cuts crew overtime & rake detention.

🌱 ENVIRONMENTAL & ENERGY IMPACT

Prevents unscheduled outer-signal stops of 24-coach rakes, saving 150–200 kWh power / 35–50L diesel per stop.

Self-Growing DB + Manual Accident Override Included

How DARPAN Beats Existing Railway Formulas, Consumer Apps & Academic Models

System / Baseline	Arrival Window (P10/P50/P90)	Fog & Track-Heat Speed Rules	Overtaking & Domino Delay	Zero-Clash Platform & Outer Hold	Self-Growing DB & Manual Accident Ops	Typical Error (MedAE) & Score
Old Railway Formula (STA + Delay – Buffer)	✗ Single Time	✗ None	✗ Isolated	✗ None	✗ Static Table	4.93m (1/6) 🚫
Consumer Apps (Where Is My Train / NTES)	~ Single Time	✗ None	✗ Isolated	✗ Unaware	✗ Read-Only App	Linear (1.5/6) 🚫
Academic GNN (IIT-KGP RSTGCN, 2024)	✗ Hourly Stn Avg	✗ No Weather	~ Station Graph	✗ No Platforms	✗ Frozen CSV	Stn Avg (2/6) 🚫
Other SIH Repos (Simulated / Basic ML)	~ Synthetic CI	✗ Month Lookup	✗ No Overtaking	✗ Platform Clashes	✗ No Feedback DB	Synthetic (2.5/6) 🚫
DARPAN (Team Twarit — SIH26028) 🏆 BEST 6/6	✓ P10 / P50 / P90	✓ 3.19M Wx + G&SR	✓ COA + FIFO Graph	✓ 704 Cabins + Tree	✓ Auto-Scrape + Ops	1.49m (6/6) 🟢 🏆

VERIFIED RAILWAY AUDITS, PAPERS & CLICKABLE SOURCES

1. Indian Railways CAG Audit & Official Operating Rules (G&SR)

Govt Audits

- **CAG Report No. 22 of 2021 (Para 2.1):** Proves fixed Engineering & Traffic Recovery buffers fail on >100% congested High-Density Network (HDN) corridors.
- **Indian Railways G&SR Rule 3.61 & Track Manual Para 5.2:** Official Fog-Pass GPS speed caps (75 km/h with FPD vs 60 km/h standard) & 60°C track-heat limits.

Sources: cag.gov.in · indianrailways.gov.in · cris.org.in

2. Machine Learning, Arrival Window Calibration & Graph Routing

Peer-Reviewed

- **IIT Kharagpur Railway Graph Study (Chowdhury et al., IEEE T-ITS 2024):** 'RSTGCN: Spatio-Temporal Graph Convolutional Network' across 4,735 Indian stations.
- **LightGBM Quantile GBDT (NeurIPS 2017) & Conformal Quantile Regression (NeurIPS 2019):** Calibrated P10/P50/P90 windows + Yen's 25kV Detours (1971).

Sources: arxiv.org/abs/2510.01262 · arxiv.org/abs/1905.03222

3. Self-Growing 6-DB Data Lake (1.68 GB) & Open Datasets

Live & Growing

- **DataMeet Indian Railways Topology + stations.db:** 8,990 stations, 420K halts, 704 cabins + Renamed-Station Code Translator (**72,508 records recovered**).
- **historical.db** (1.51M delays, 73,342 slack profiles) + Open-Meteo Weather Grid (3.19M rows) + 24x7 SnapshotWriter feedback loop.

Sources: github.com/datameet/railways · open-meteo.com

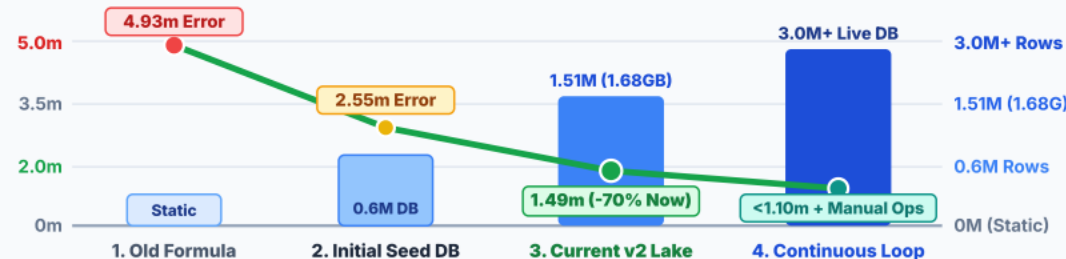
SCAN QR OR CLICK LINKS TO TEST LIVE PROTOTYPE

- Live Web Portal: <https://team-twarit-sih26028.vercel.app>
- Demo Video: [Google Drive Walkthrough](#)
- Instant API & Interactive Platform Maps across 17 Zones.

VISUAL ANALYTICS: SELF-GROWING DB FLYWHEEL & DELAY DRIVERS

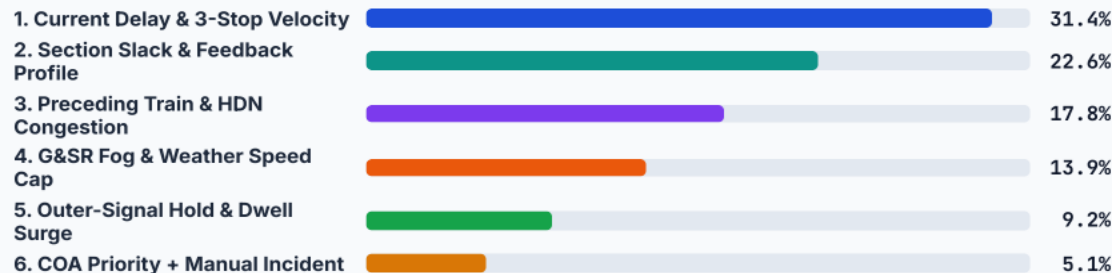
Graph 1: Self-Growing DB & Feedback Flywheel (Longer It Runs → Lower Error)

-69.8% Error



Graph 2: What Drives Train Delays? (Learned AI Features + Manual Ops)

31 Features



1.68 GB Self-Growing 6-Database Distribution (Grows 24x7 via Live Scraper): 85.3% Window Coverage



[Click on blue texts to open link] — Every research study (IIT Kharagpur RSTGCN, NeurIPS), CAG Audit & Live Demo URL above is directly clickable

[Open Live Portal ↗](#)